

Use-Case Flow Specification

Problem Statement #39 Second-Hand Product Escrow Marketplace

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Section: 5C

UC-07: Approve Product & Release Escrow Funds

Primary Actor	Buyer
Secondary Actors	System (Escrow Engine), Inspection Agent (report already on file)
Trigger	Buyer opens the order after being notified that the inspection report has been posted.

Preconditions

- The buyer has an authenticated, active account and valid order in state called Under Inspection.
- An inspection agent has submitted a physical verification report for the order.
- Buyer payment for the order is held in escrow (which is locked and not yet released to the seller).

Postconditions

- Success: Order status is 'Completed' and escrow funds are released to the seller; buyers and seller both receive a closure notification and receipt.
- Failure/Alternate: Order status is 'Disputed' and escrow funds remain frozen; case is queued for admin review.

Main Success Scenario

1. Buyer selects order from “My Orders” and opens the posted inspection report (photos, condition notes, pass/fail checklist).
2. System displays report alongside the original listing details and an “Approve” / “Reject” action pair, with visible 48-hour countdown for the review window.
3. Buyer reviews report and selects “Approve.”
4. System prompts buyer to confirm the decision, noting that approval is the final and will trigger fund release.
5. Buyer confirms the approval.
6. System validates that the order is still in state 'Under Inspection' and within the 48-hour window.
7. System transitions the order state to 'Completed' and instructs the escrow engine to release the held funds to the seller's payout account.
8. System logs the release transaction with a timestamp for audit purposes.
9. System sends a confirmation notification and digital receipt to both buyer and seller.
10. Use case ends successfully.

Alternate Flow

6a. Buyer Rejects the Product / Files a Dispute (branches at step 3):

- 6a1. Buyer selects “Reject” instead of “Approve” and enters a reason (for e.g., item condition does not match listing or inspection report).
- 6a2. System immediately transitions the order to 'Disputed' status and freezes the escrow funds, blocking any scheduled release (per FR-005 / NFR-001).
- 6a3. System opens a dispute ticket, attaches the buyer's reason and the inspection report, and then routes it to an Admin for resolution (UC-09).
- 6a4. Buyer and seller are both notified that the order is under dispute review.
- 6a5. Use case ends with the order pending admin resolution rather than automatic completion.

9a. Review Window Expires Without a Decision (branches at step 2):

- 9a1. If the buyer takes no action within 48 hours of the report being posted, the system will automatically flag the order for admin review (per FR-004).
- 9a2. Admin reviews the case and either releases funds to the seller or opens a dispute, resuming main flow at step 7 or the 6a branch accordingly.